

00:56

Inference Rules in Propositional Logic Artificial Intelligence

- Modes Ponens (MP) (Implication-elimination)
- Example:
 - If it is raining, then Rahul is inside
 - It is raining
 - $\alpha - It\ is\ raining$
 - $\beta - Rahul\ is\ inside$

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01:43

03:02

Inference Rules in Propositional Logic Artificial Intelligence

- Modes Tollens (MT)
- Example:
 - If it is raining, then Rahul is inside
 - Rahul is not inside
 - $\alpha - It\ is\ raining$
 - $\beta - Rahul\ is\ inside$
 - $\alpha \rightarrow \beta$

$$\frac{\alpha \Rightarrow \beta, \neg \beta}{\neg \alpha}$$

- $\neg\beta$
- $\therefore \neg\alpha$ ✓

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03:37

Inference Rules in Propositional Logic Artificial Intelligence

- And Elimination (AE)
- Example:
- Sachin and Sourav are friends of Rahul ✓
- α – Sachin is a friend of Rahul
- β – Sourav is a friend of Rahul
- $\alpha \wedge \beta$

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03:39

Inference Rules in Propositional Logic Artificial Intelligence

- And Elimination (AE)
- Example:
- Sachin and Sourav are friends of Rahul ✓
- α – Sachin is a friend of Rahul
- β – Sourav is a friend of Rahul

- $\alpha \wedge \beta$

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03:39

Inference Rules in Propositional Logic Artificial Intelligence

- And Elimination (AE)
- Example:
 - Sachin and Sourav are friends of Rahul ✓
 - α – Sachin is a friend of Rahul
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 - $\alpha \wedge \beta$

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05:25

Inference Rules in Propositional Logic Artificial Intelligence

- AND Introduction (AI)
- Example:
 - Rahul pass the exam ✓
 - Sachin pass the exam ✓
 - α – Rahul pass the exam

$$\alpha_1, \alpha_2, \dots, \alpha_n$$

α – Rahul pass the exam

- β – Sachin pass the exam
- α ✓
- β ✓
- $\therefore \alpha \wedge \beta$

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}{-}$$

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05:26

Inference Rules in Propositional Logic Artificial Intelligence

- AND Introduction (AI)
- **Example:**
- Rahul pass the exam ✓
- Sachin pass the exam ✓
- α – Rahul pass the exam
- β – Sachin pass the exam
- α ✓
- β ✓
- $\therefore \alpha \wedge \beta$

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}$$

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Inference Rules in Propositional Logic Artificial Intelligence

- **Modes Ponens (MP) (Implication-elimination)**
- **Example:**
- If it is raining, then Rahul is inside

- It is raining
- α – *It is raining*
- β – *Rahul is inside*
- $\alpha \rightarrow \beta$
- α
- $\therefore \beta$

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05:45

Inference Rules in Propositional Logic Artificial Intelligence

- OR Introduction (AI)
- Example:
- Rahul pass the exam ✓
- α – Rahul pass the exam •

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06:01

Inference Rules in Propositional Logic Artificial Intelligence

- OR Introduction (AI)

- **Example:**

- Rahul pass the exam ✓

- α – Rahul pass the exam

- α ✓

- $\therefore \underline{\alpha \vee \beta}$ ✓



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06:03

Inference Rules in Propositional Logic Artificial Intelligence

- **OR Introduction (AI)**

- **Example:**

- Rahul pass the exam ✓

- α – Rahul pass the exam

- α ✓

- $\therefore \underline{\alpha \vee \beta}$ ✓



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06:23

Inference Rules in Propositional Logic Artificial Intelligence

- **Negation Elimination (NE) or Double Negation**
- **Example:**
- It is not true that Rahul did not pass the exam

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07:45

Inference Rules in Propositional Logic Artificial Intelligence

- **Unit Resolution (UR)** ✓
- **Example:**
- Rahul or Sachin pass the exam ✓
- Sachin did not pass the exam ✓
- α – Rahul pass the exam
- β – Sachin pass the exam

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07:57

Inference Rules in Propositional Logic Artificial Intelligence

- Unit Resolution (UR) ✓
- Example:
 - Rahul or Sachin pass the exam ✓
 - Sachin did not pass the exam ✓
 - α – Rahul pass the exam
 - β – Sachin pass the exam
 - $\alpha \vee \beta$ ✓
 - $\neg\beta$ ✓

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08:16

Inference Rules in Propositional Logic Artificial Intelligence

- Unit Resolution (UR) ✓
- Example:
 - Rahul or Sachin pass the exam ✓
 - Sachin did not pass the exam ✓
 - α – Rahul pass the exam
 - β – Sachin pass the exam
 - $\alpha \vee \beta$ ✓
 - $\neg\beta$ ✗
 - $\therefore \alpha$

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Inference Rules in Propositional Logic Artificial Intelligence

- Unit Resolution (UR) ✓
- Example:
- Rahul or Sachin pass the exam ✓
- Sachin did not pass the exam ✓
- α – Rahul pass the exam
- β – Sachin pass the exam
- $\alpha \vee \beta$ ✓
- $\neg\beta$ ✗
- $\therefore \alpha$

$$\frac{\alpha \vee \beta, \quad \neg\beta}{\alpha}$$

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Inference Rules in Propositional Logic Artificial Intelligence

- Unit Resolution (UR) ✓
- Example:
- Rahul or Sachin pass the exam ✓
- Sachin did not pass the exam ✓
- α – Rahul pass the exam
- β – Sachin pass the exam
- $\alpha \vee \beta$ ✓
- $\neg\beta$ ✗
- $\therefore \alpha$

$$\frac{\alpha \vee \beta, \quad \neg\beta}{\alpha}$$

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